

Nandakumar Vuppalapati

AI Data Engineer — Python, AWS, Databricks, dbt, Spark

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PROFESSIONAL SUMMARY

Data Engineer building production ETL/ELT pipelines and lakehouse platforms for regulated financial-services and healthcare data, across internship, co-op, and full-time roles. Comfortable owning workflows end to end across AWS and Azure — from ingestion and dimensional modeling through testing, observability, and production support — with recent work extending into ML feature pipelines and LLM/RAG tooling for compliance and document-search use cases.

TECHNICAL SKILLS

Languages: Python (Pandas, PySpark), SQL (window functions, CTEs, query tuning), T-SQL, Bash

Cloud: AWS (S3, Glue, Lambda, Step Functions, Redshift, Athena), Azure (Databricks, Data Factory, Synapse Analytics, Blob Storage/ADLS Gen2, Azure SQL)

Data Processing: Apache Spark, PySpark, Databricks, Spark Structured Streaming, Apache Kafka

Orchestration & Transformation: Apache Airflow (DAGs, scheduling, sensors, retries, failure handling), dbt, Azure Data Factory

Warehouse / Lakehouse: Snowflake, Databricks, Delta Lake, AWS Redshift, Azure Synapse; Medallion Architecture (Bronze/Silver/Gold)

Databases: Oracle, PostgreSQL, MySQL, Azure SQL, Databricks SQL, Pinecone (vector database)

DevOps / IaC: Docker, Terraform, GitHub Actions (CI/CD), Git

Data Reliability & Governance: Great Expectations, dbt tests, data contracts, data lineage, PII masking, RBAC, SOX-aware controls, compliance documentation

AI / ML Data Systems: MLflow, feature engineering, XGBoost, RAG, vector retrieval, LangChain, OpenAI API, Pinecone, Feast, Ragas, FastAPI

WORK EXPERIENCE

AI Data Engineer · Workiva, Remote, USA

Dec 2025 – Present

- Own Python/Airflow pipelines across AWS S3, Glue, Lambda, Step Functions, and Redshift, processing **1M+ daily records** across payments, risk, settlement, and compliance; Prometheus/Grafana monitoring traces failures and validates recovery before pipelines resume normal schedules.
- Architected Snowflake/dbt transformation layers with reusable SQL models, macros, and automated tests, maintaining **30+ financial models** that publish governed datasets for reporting and risk analysis.
- Delivered Spark/Databricks feature and embedding pipelines over curated financial data for downstream fraud and ML workflows, supporting feature preparation and model lifecycle with MLflow.
- Prototyped a regulatory-document RAG proof of concept with LangChain and OpenAI APIs, combining chunking, embeddings, and retrieval so compliance users could search regulatory content in natural language.

Data Engineer (Co-op) · Wellmark BCBS, Des Moines, USA

Aug 2024 – May 2025

- Engineered Azure Data Factory ingestion pipelines consolidating **10M+ records** from 30+ legacy healthcare sources into Blob Storage and Azure SQL, applying PII controls and tracking lineage via Terraform, Docker, and GitHub Actions.
- Modernized PySpark transformation jobs in Azure Databricks to clean, standardize, join, and curate data across medallion-style layers, replacing legacy SQL-heavy processing.
- Authored **15+ SQL/dbt transformation models** with schema tests and Great Expectations checkpoints, catching schema, null, duplicate, and integrity issues before curated data reached downstream consumers.
- Developed an LLM-assisted search prototype across a **1,000+ document** vendor-contract repository using document extraction, embeddings, and vector retrieval for natural-language Q&A.

Junior Data Engineer · Sigmoid, Bengaluru, India

Jun 2022 – Dec 2023

- Refactored T-SQL stored procedures and indexing strategies behind recurring finance-reporting workloads, resolving long-running queries and month-end reporting bottlenecks.
- Designed a Python-based validation framework with pytest checks integrated into ETL processing, catching schema and null-value issues before data reached stakeholder reports.
- Redesigned a flat reporting structure into a star-schema dimensional model in Azure Synapse; automated a manual Excel-based weekly workflow with scheduled ADF pipelines, saving roughly **6 hours** of manual work weekly.
- Diagnosed and resolved failed ADF production pipeline runs, validated outputs after recovery, configured schedule triggers and failure alerts, and managed Blob Storage lifecycle policies — handling reruns for finance analysts when upstream data arrived late.

Data Scientist (Intern) · Tech Citi Software, Bengaluru, India

Jul 2021 – Dec 2021

- Migrated historical loan and transaction data from Oracle to PostgreSQL, resolving schema and datatype mismatches and validating migrated records for consistent downstream reporting.
- Assembled Python and Apache Airflow ETL workflows to ingest **50,000+ daily** financial transaction records from CSV and REST API sources into PostgreSQL, replacing manual data-loading steps.
- Engineered transaction-level behavioral features and trained an offline XGBoost fraud-classification prototype with SMOTE, reaching an **AUC-ROC of 0.85** in historical-data evaluation.

PROJECTS

Real-Time ML Feature Platform for Fraud Scoring

GitHub 

Kafka, Flink, Feast, Databricks, MLflow, FastAPI, Docker, AWS

- Built to close the gap between batch fraud pipelines and low-latency scoring — Kafka handles event ingestion, Flink computes streaming features, Feast aligns online/offline stores, and MLflow/FastAPI handle versioning and serving, with each layer containerized as a separate, independently evolvable concern.

CDC Lakehouse with Data Contracts

GitHub 

Debezium, Kafka, Apache Iceberg, dbt Core, Great Expectations, PostgreSQL, Docker

- Debezium captures PostgreSQL row-level changes, Kafka transports them into Apache Iceberg, and dbt plus Great Expectations enforce schema-compatibility contracts so breaking source changes fail visibly instead of silently propagating downstream.

RAG Evaluation & Observability Harness

GitHub 

LangChain, OpenAI API, Pinecone, Ragas, FastAPI, Docker, Airflow

- A golden-question dataset evaluates retrieval and generation separately, with Ragas and scheduled Airflow runs surfacing hallucination and retrieval regressions.

EDUCATION & CERTIFICATIONS

MS, Computer and Information Sciences — University of South Dakota, 2025 | B.Tech — IIIT, India, 2023

AWS Certified Cloud Practitioner — Amazon Web Services, 2025